

Final Project 270C: Optimal Control of a Variable-Mass Thrust-Vectoring Lander

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1 Overview

This project studies the optimal guidance and control of a planar variable-mass rocket/lander with translational and rotational dynamics, bounded controls, nonlinear coupling, free-final-time optimal control, endpoint manifold constraints, and trajectory tracking. The project combines Pontryagin's Maximum Principle (PMP), controllability analysis, Linear Quadratic Regulation (LQR), minimum-time optimal control, nonlinear indirect optimal control, and numerical two-point boundary value problem (TPBVP) solvers.

The project is inspired by powered-descent guidance and reusable rocket landing systems studied in modern aerospace optimal control literature. All theoretical derivations must be based on PMP. Numerical implementations may use indirect shooting, multiple shooting, continuation or homotopy methods, and indirect/direct hybrid approaches, provided that the optimality conditions are derived from PMP.

The project consists of four parts:

In Part I, you will study the linearized dynamics with a quadratic cost and solve a fixed-time, free-endpoint optimal control problem. Since the resulting TPBVP can be analytically reduced to a Riccati differential equation, this part allows you to efficiently compute the optimal solution while developing intuition for LQR structure, controllability, and the effect of weighting matrices on closed-loop behavior.

In Part II, you will extend the LQR framework to trajectory tracking by deriving and implementing the optimal controller for following a prescribed reference trajectory. You are free to select a physically reasonable reference trajectory for illustration and analysis.

In Part III, you will solve a complete mission problem consisting of two phases using the linearized dynamics: a minimum-time ascent phase to a prescribed point in the sky, followed by a free-final-time landing phase with terminal manifold constraints.

Finally, in Part IV, you will work with the full nonlinear variable-mass dynamics. To keep the nonlinear problem computationally manageable, this part focuses on the simpler fixed-time, free-endpoint optimal control problem, allowing you to study the effect of nonlinear coupling and mass depletion while comparing linear and nonlinear optimal solutions.

2 Overall Expectations

This project is intended to be mathematically rigorous and computationally substantial. Your final submission must include a complete written report, fully runnable simulation code, clear derivations, numerical validation, figures and illustrations, and interpretation of results.

2.1 Modular Software Requirement

Your software implementation must be modular and configurable. The user should be able to modify physical parameters, cost matrices, terminal conditions, thrust bounds, target sets, and model selections without rewriting major portions of the code. At minimum, your implementation should separate dynamics, optimal control solvers, numerical integrators, plotting utilities, simulation configuration, and experiment scripts. The code should support parameter variation over a reasonable operating range while maintaining numerical stability.

2.2 Use of Generative AI for Coding:

You are **encouraged to use generative AI coding tools** as development aids. However, you are fully responsible for the correctness, structure, and behavior of your final implementation. You must be able to explain, derive, debug, and modify every part of the submitted code. **Using AI-generated code without full understanding is not acceptable** and may be treated as academic misconduct if you cannot explain your implementation during project discussions or evaluations.

3 Required Deliverables

1) Written Report: Your report must present a complete and coherent study of the project, including the mathematical formulation, derivations, assumptions, controllability analysis, numerical methods, simulation results, validation, discussion, and limitations. Every section must include clear figures and illustrations together with interpretation of the corresponding physical, mathematical, and numerical behavior. Merely presenting plots or numerical outputs without analysis will receive little credit.

The report should be written in a professional scientific style with sufficient detail for reproducibility. You are expected to justify modeling assumptions, explain numerical choices, discuss convergence behavior and numerical sensitivity, and critically evaluate the validity and limitations of your results.

Use of Generative AI for Writing: You may use generative AI tools only for editing, grammar correction, formatting assistance, or improving clarity of writing. The technical content, derivations, explanations, analysis, and interpretation **must be entirely your own work**. Submitting AI-generated technical explanations or derivations that you do not fully understand constitutes academic misconduct.

2) Code Submission: You must submit fully runnable code in MATLAB, Python, Julia, or C++. The code must reproduce all figures and numerical results presented in the report, include execution instructions, and contain sufficient comments and documentation to allow another user to understand and run the implementation.

Your software must be modular and configurable. The user should be able to modify physical parameters, cost matrices, target sets, terminal conditions, and solver settings without rewriting major portions of the code.

4 Physical System

Consider the planar (2D) variable-mass rocket/lander with state

$$x = [p_x \ p_z \ v_x \ v_z \ \theta \ \omega \ m]^\top,$$

where (p_x, p_z) denote position, (v_x, v_z) denote velocity, θ is the pitch angle, ω is the angular velocity, and m is the vehicle mass. The control input is

$$u = \begin{bmatrix} T \\ \tau \end{bmatrix},$$

where T is the thrust magnitude and τ is the control torque.

4.1 Nonlinear Dynamics

The nonlinear dynamics are given by

$$\begin{aligned} \dot{p}_x &= v_x, & \dot{p}_z &= v_z, & \dot{v}_x &= \frac{T}{m} \sin \theta, & \dot{v}_z &= \frac{T}{m} \cos \theta - g, \\ \dot{\theta} &= \omega, & \dot{\omega} &= \frac{\tau}{I}, & \dot{m} &= -\alpha T, \end{aligned}$$

where $g > 0$ is gravity, $I > 0$ is the moment of inertia, and $\alpha > 0$ is the energy consumption coefficient.

4.2 Control Constraints

Assume the control constraints

$$T \in [T_{\min}, T_{\max}], \quad \tau \in [\tau_{\min}, \tau_{\max}].$$

Optional: You may additionally impose tilt constraints, angular-rate constraints, glide-slope constraints, or thrust-pointing constraints if they are physically motivated and properly justified.

Note: You are responsible for selecting physically reasonable parameters and mission configurations that preserve feasibility of the optimal control problems while generates interesting trajectories. **Suggested Parameter ranges:**

$$\begin{aligned} g &= 9.81 \text{ m/s}^2, & I &= 10 \text{ kg}\cdot\text{m}^2, & m_0 &= 20 \text{ kg}, & \alpha &= 0.005 \text{ kg}/(\text{N}\cdot\text{s}), \\ T_{\min} &= 0 \text{ N}, & T_{\max} &= 300 \text{ N}, & \tau_{\min} &= -30 \text{ N}\cdot\text{m}, & \tau_{\max} &= 30 \text{ N}\cdot\text{m}. \end{aligned}$$

You can change these parameters to improve the feasibility of the problem (like reducing α or increasing $T_{\max}$) as long as it is within a reasonable range.

4.3 Numerically Enforced State Constraints

The physically meaningful state constraints are

$$p_z(t) \geq 0, \quad m(t) \geq m_{\text{dry}} > 0,$$

for all admissible trajectories. The altitude constraint prevents the vehicle from penetrating the ground before landing, while the mass constraint prevents unphysical negative-mass or below the dry mass trajectories caused by excessive energy consumption.

It becomes much harder to enforce these as state constraints for the optimal control problem. So, you are not required to enforce these constraints through a full constrained optimal control formulation. However, your chosen parameters, trajectories, and numerical solutions must satisfy these constraints throughout the simulations.

5 Part I — Fixed-Time LQR Around Different Nominal Trajectories

5.1 Linearization Around Hover

Linearize the dynamics around the hover equilibrium

$$v_x^* = v_z^* = \theta^* = \omega^* = 0, \quad T^* = mg.$$

Define the thrust deviation input

$$\delta T = T - mg, \quad u = \begin{bmatrix} \delta T \\ \tau \end{bmatrix},$$

and derive the linearized dynamics

$$\dot{x} = Ax + Bu.$$

5.2 Weakly Controllable Linearization

Construct a second linearization around the nominal descending trajectory

$$\theta^* \approx 0, \quad v_x^* \gg 0, \quad v_z^* < 0, \quad T^* \approx mg,$$

corresponding to a nearly vertical powered descent with large horizontal velocity and thrust operating close to hover thrust. In this regime, horizontal acceleration is generated primarily through attitude deviations, producing weak coupling between translational and rotational dynamics and leading to poor controllability in the horizontal direction.

Linearize the dynamics along this trajectory and:

1. derive the resulting time-varying linearized dynamics,
2. compute controllability matrices and finite-horizon controllability Gramians,
3. compare the smallest singular values/eigenvalues with those of the hover linearization,
4. and explain physically why controllability deteriorates along this trajectory.

5.3 Optimal Control Problem

For both linearizations, solve

$$\min_u \int_0^{t_f} \left(x^\top Q x + u^\top R u \right) dt + x(t_f)^\top Q_f x(t_f)$$

with fixed final time and free terminal state.

5.4 Required Derivations

Derive the PMP conditions, adjoint equations, stationarity conditions, Riccati differential equation, and optimal state-feedback law.

5.5 Required Numerical Analysis

First, compare multiple choices of Q and R and Q_f matrices to improve the performance. You can choose these matrices to be diagonal with diagonal elements of different magnitude. The main purpose of this part is to find the right parameter range to induce the “good” trajectories. If you, instead, normalize the dynamics by working with nondimensional quantities (through scaling each state against a “good” reference value) then a natural choice of these matrices would be just identity matrices.

Second, analyze the differences between the controllable and weakly controllable linearizations by comparing the sensitivity to initial condition error, the control energy consumed, and the transient behavior.

6 Part II — Optimal Trajectory Tracking

Consider the linearized dynamics derived in Part I and record one of the optimal state trajectories generated in Part I as the reference trajectory

$$x_{\text{ref}}(t).$$

Instead of initializing the system near this reference trajectory, choose a substantially different initial condition and design an optimal tracking controller capable of driving the system toward the reference trajectory while minimizing tracking error and control effort. This resembles scenarios where the vehicle arrives at a new phase of a mission with some error in the initial condition but has to eventually track the previously planned reference trajectory for the subsequent missions.

Solve the following finite-horizon tracking problem

$$\min_u \int_0^{t_f} \left[(x - x_{\text{ref}})^\top Q (x - x_{\text{ref}}) + u^\top R u \right] dt.$$

The selected initial condition should be sufficiently different from the reference trajectory to produce nontrivial transient behavior and meaningful tracking dynamics.

6.1 Required Derivations and Analysis

Derive the PMP conditions, adjoint equations, stationarity conditions, and the resulting tracking Riccati equations. Compare the trajectory tracking problem with the regulation problem studied in Part I, and analyze the sensitivity of the closed-loop solution to perturbations in the initial condition and reference trajectory. Your analysis should include:

- tracking performance,
- transient response,
- control effort,
- and robustness to deviations from the nominal trajectory.
- Discuss the following question: *What if the reference trajectory is not known in advance?* Your discussion must explain what challenges does your current approach face and why.

7 Part III — The Mission Problem

The mission consists of two phases: a minimum-time ascent phase to a predefined state x_{sky} , followed by a return-and-land phase. Note that you need to choose x_{sky} reasonably for feasibility of an optimal solution.

7.1 Phase A — Minimum-Time Ascent

Starting from a prescribed initial condition, drive the vehicle to a fixed rendezvous point in the sky in minimum time. The optimal control problem is

$$\min_u J = t_{f_1}$$

subject to the system dynamics, bounded controls, and terminal condition

$$x(t_{f_1}) = x_{\text{sky}}.$$

You must derive the PMP conditions, switching functions, and characterize the structure of the optimal control, including the existence or nonexistence of singular arcs. Determine whether the optimal solution exhibits bang-bang behavior and solve the resulting TPBVP numerically.

Your analysis should explain why thrust saturation appears, how mass depletion changes control authority during flight, how the switching structure evolves over time, and whether the optimal strategy front-loads or back-loads energy consumption.

Note that the behavior largely depends on the choice of x_{sky} . Choose the terminal velocity to be zero versus it is positive/negative and compare.

7.2 Phase B — Return-and-Land Problem

Starting from the sky rendezvous point obtained in Phase A, the vehicle must return and land while minimizing the input thrust energy

$$J = \int_{t_{f_1}}^{t_{f_2}} (u^\top R u) dt,$$

with free final time t_{f_2} . Two terminal manifolds must be considered.

Landing Manifold 1 — Ground Landing:

$$\mathcal{M}_1 = \{p_z = 0, v_z = 0, \theta = 0\},$$

corresponding to landing on flat ground with vertical touchdown and upright attitude.

Landing Manifold 2 — Circular Platform: Define

$$h_1(x) = (p_x - p_c)^2 + p_z^2 - r^2,$$

together with

$$v_z = 0, \quad \theta = 0,$$

so that

$$\mathcal{M}_2 = \{h_1(x) = 0, v_z = 0, \theta = 0\}.$$

7.3 Required Derivations and Analysis

For both landing manifolds, derive the PMP conditions, transversality conditions, manifold end-point conditions, free-final-time conditions, and the Hamiltonian terminal condition

$$H(t_f) = 0.$$

Compare the two landing scenarios in terms of landing time, energy consumption, switching structure, terminal sensitivity, and feasibility. Your discussion should explain how the geometry of the terminal manifold affects the resulting optimal trajectory and control structure.

8 Part IV — The Nonlinear Optimal Control Near Hover

Return to the full nonlinear variable-mass model and restrict attention to near-hover trajectories, moderate attitude angles, and short maneuver durations. The goal of this section is to study the effect of nonlinear coupling and mass depletion, and to determine the validity region of the linearized optimal control solutions obtained in previous sections.

Unlike Part III, you are **not** required to solve a fully nonlinear free-final-time minimum-time problem. Instead, consider the fixed-time, free-endpoint nonlinear optimal control problem

$$\min_u \int_0^{t_f} (x^\top Qx + u^\top Ru) dt,$$

subject to the full nonlinear dynamics and bounded controls.

8.1 Required Derivations

Derive the nonlinear Hamiltonian, adjoint equations, stationarity conditions, and the resulting TPBVP. Clearly identify the state equations, costate equations, terminal conditions, and optimality conditions.

8.2 Linear vs Nonlinear Comparison

Using the same initial condition, simulate:

1. the linear optimal controller on the linearized dynamics,
2. the same controller on the nonlinear dynamics,
3. and the nonlinear optimal controller on the nonlinear dynamics.

Compare trajectories, terminal behavior, energy consumption, and control effort. Your discussion should explain when the linear approximation remains accurate, when it breaks down, and how nonlinear translational–rotational coupling and mass depletion affect the resulting trajectories.

8.3 Numerical Solution and Analysis

Solve one nonlinear optimal control problem numerically using indirect shooting, multiple shooting, or collocation. The nonlinear solver must be initialized using the corresponding linearized optimal solution.

Your numerical analysis must include trajectory comparisons, control histories, Hamiltonian evolution, convergence behavior, solver sensitivity, and terminal residual analysis. You should also

discuss the numerical sensitivity of the nonlinear TPBVP and the effect of nonlinear coupling on the optimal solution structure.

8.4 Required Figures

At minimum, include position, velocity, attitude, and mass trajectories together with thrust and torque histories, Hamiltonian evolution, and terminal error comparisons. Every figure must be accompanied by explanation and interpretation.

9 Validation Requirements

All numerical solutions must be carefully validated. At minimum, you must verify terminal conditions, Hamiltonian consistency, adjoint boundary conditions, and shooting residuals associated with the TPBVP formulations. Simply presenting trajectories or plots is insufficient; you must demonstrate that the computed solutions satisfy the necessary optimality conditions to acceptable numerical tolerance.

Note: In all of the sections, in addition to illustration of the solution trajectories vs time, you should also provide **the mission trajectory** that plots p_x vs p_z and also a small vector denoting the attitude at various positions.

10 Suggested Numerical Tools

You may use any suitable scientific computing environment. Commonly used tools include MATLAB (`bvp4c`, and sometimes `fmincon`), CasADi, IPOPT, GPOPS-II, SciPy, Julia `OptimalControl.jl`, or custom indirect shooting implementations.

11 Grading

Derivations and Theory	35%
Numerical Implementation	35%
Interpretation and Discussion	20%
Presentation Quality	10%

12 Academic Expectations

You are expected to justify modeling assumptions, explain numerical methods and solver choices, discuss convergence behavior and numerical sensitivity, identify limitations of your approach, and interpret the physical meaning of the resulting trajectories and controls. Simply producing numerical trajectories without analysis or interpretation will receive little credit.

13 References and Background Reading

You are encouraged to consult standard references in optimal control, including Liberzon's *Calculus of Variations and Optimal Control Theory*, Bryson and Ho's *Applied Optimal Control*, Kirk's *Optimal Control Theory: An Introduction*, together with modern powered-descent guidance and trajectory optimization literature.